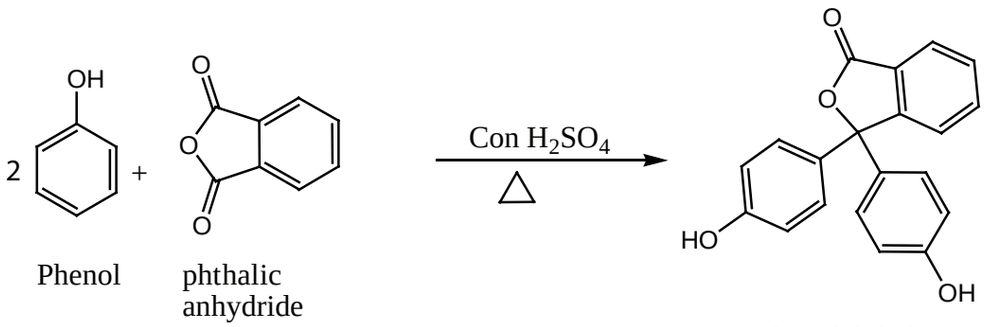
vii) Phthalein reaction:

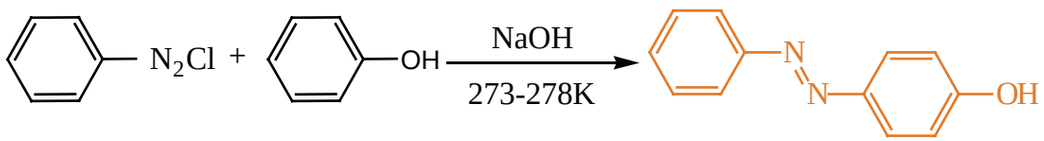
On heating phenol with phthalic anhydride in presence of con.H_2SO_4, phenolphthalein is obtained.



Phenol | phthalic anhydride | Phenolphthalein

viii) Coupling reaction:

Phenol couples with benzene diazonium chloride in an alkaline solution to form p-hydroxy azobenzene(a red orange dye).



Benzene diazonium chloride | Phenol | p-hydroxy azobenzene

Test to differentiate alcohol and phenols

i) Phenol react with benzene diazonium chloride to form a red orange dye, but ethanol has no reaction with it.

ii) Phenol gives purple colouration with neutral ferric chloride solution, alcohols do not give such coloration with $FeCl_3$.

iii) Phenol reacts with NaOH to give sodium phenoxide. Ethyl alcohol does not react with NaOH.

Uses of phenol

1) About half of world production of phenol is used for making phenol formaldehyde resin. (Bakelite).
2) Phenol is a starting material for the preparation of
   i) drugs such as phenacetin, Salol, aspirin, etc.
   ii) phenolphthalein indicator.
   iii) explosive like picric acid.
3) It is used as an antiseptic-carbolic lotion and carbolic soaps.

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